

TEST-04-CHARLEMAGNE-FLEET-TESTING

TEST-04: CHARLEMAGNE FLEET TESTING PROTOCOL

Carrington Storm Motors — Project AEGIS

Hull GIC Susceptibility • Amphibious Mechanism • Electronics Hardening • Propulsion • Safety

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TESTING PHILOSOPHY

A cruise vessel is a floating city of 3,000 souls, 100,000 gross tonnes, and 1,000 feet of steel hull that acts as a geomagnetic antenna. The Charlemagne testing protocol must verify that CSM’s hardening transforms this 1,000-foot antenna into a 1,000-foot insulator — reducing hull-induced current to <0.03% of unprotected baseline. Testing spans three environments: dry dock (controlled current injection), sea trials (simulated GIC in open water), and in-service monitoring (continuous data during commercial operation).

PHASE 1: COMPONENT-LEVEL VALIDATION (TRL 3→4→5)

1.1 MXene Hull Coating — Marine Environment

Test	Method	Specimen	Duration	Pass/Fail
SE on Steel Substrate	ASTM D4935 (TEM cell) — MXene on shipbuilding steel (AH36 grade), 300×300×10 mm plates with surface preparation per marine coating standard (SSPC-SP10 near-white blast)	5 coated plates, 5 uncoated control	—	SE >85 dB at 10 MHz–1 GHz on steel (steel’s own SE is reflection-based; MXene adds absorption)
Salt Spray	ASTM B117, 5% NaCl, 35°C	10 coated plates	1,000 hours (extended from standard 500 — marine requirement)	SE degradation <5 dB; no blistering per ASTM D714
Cyclic Corrosion (CCT)	ISO 20340 (Procedure A for offshore — 72h UV/condensation + 72h salt spray + 24h low-temperature at −20°C)	10 coated plates	25 cycles (4,200 hours)	SE degradation <10 dB; no under-film corrosion >2 mm from scribe
Cathodic Disbondment	ASTM G8 (impressed current cathodic protection at −1.5V	5 coated plates	28 days	Disbondment radius <10 mm from intentional holiday;

Antifouling Compatibility	vs. SCE for 28 days in synthetic seawater)		coating remains adherent
	Apply commercial antifouling (silyl acrylate self- polishing) over MXene; verify no SE degradation and no MXene delamination	5 plates with MXene + antifouling topcoat	6 months seawater immersion (static raft test)
			SE within 5 dB; no delamination; antifouling performance equivalent to standard hull coating

Cost: \$124K; Duration: 12 months (cyclic corrosion is longest-lead test)

1.2 BFRP Structural Elements — Marine Grade

Test	Method	Acceptance
Tensile	ASTM D3039 (direction 0° and 90°)	>1,000 MPa (saltwater-saturated condition)
Compression	ASTM D6641	>600 MPa
Flexural	ASTM D7264	>800 MPa
Interlaminar Shear	ASTM D2344	>50 MPa
Water Absorption	ASTM D570 (24h immersion)	<0.5% weight gain
Fire	IMO FTP Code Part 1 (non-combustibility), Part 5 (surface flammability)	Pass IMO criteria for non-combustible material; surface flammability within SOLAS limits
Smoke & Toxicity	IMO FTP Code Part 2	Pass IMO criteria

1.3 PMMA POF Wiring — Marine Installation

Test	Method	Acceptance
Tensile (cable)	IEC 60794-1-2 (optical fiber cable tensile)	>500 N without optical performance degradation
Bend Radius	OTDR measurement at minimum bend	<30 mm minimum bend radius without insertion loss increase >0.1 dB
Fire	IEC 60332-1-2 (vertical flame propagation for single cable)	Self-extinguishing within 425 mm of top clamp
Smoke	IEC 61034-2 (smoke density, 3 m cube test)	>60% light transmittance (low smoke)
Halogen-Free	IEC 60754-1 (halogen acid gas)	<0.5% HCl equivalent
Vibration	MIL-STD-167-1 (shipboard vibration, Type I — environmental)	No connector loosening; OTDR stable throughout
Water Immersion	30 days at 3 m water depth (simulated cable tray flooding)	No water ingress; OTDR stable
Data Rate — 1,000 m	Bit error rate test (BERT) at 1 Gbps, 1,000 m POF with 10 connectors	BER <10 ⁻¹²

PHASE 2: SUB-SCALE VESSEL SECTION (TRL 4→5→6)

2.1 Test Article — 10 m Hull Section

Construct a 10 m × 3 m × 3 m vessel hull section replicating: - Outer hull: MXene-coated AH36 steel, 20 mm thickness - Inner structure: BFRP stiffeners at 600 mm spacing - Deck section: BFRP grating with MXene mesh overlay - Cable tray: 1,000 m PMMA POF (representative 1/50th scale of vessel data network) - Battery room mock-up: LiFePO₄ rack at 1:10 scale (2 MWh equivalent) - Watertight door: Dielectric hinge, MXene gasket - Ballast tank section: Coated interior with impressed current CP system

2.2 Test Sequence — Hull Section

Test	Equipment	Procedure	Acceptance
Dry Dock Current Injection	500 A AC source, 0.001-1 Hz; electrode array in water tank (seawater simulant: 3.5% NaCl) surrounding hull section	Inject current into water → measure hull current distribution via 24 Rogowski coils at BFRP/steel interfaces	Hull-induced current <0.03% of injected current; BFRP/steel interface current <1 mA
Hydrostatic Pressure	Seawater tank, pressurize to 0.5 MPa (50 m depth equivalent) on hull exterior	24-hour sustained pressure; visual inspection + ultrasonic thickness measurement	No water ingress through coating; no delamination; coating thickness ±5% of pre-test
Fire Scenario	IMO FTP Code — pool fire (heptane) under hull section for 30 minutes	Temperature at inner hull surface; structural integrity post-fire	Inner hull temperature <200°C (BFRP glass transition ~140°C — must not exceed); no structural collapse

Phase 2 Cost: \$1.8M (10 m hull section construction + testing); Duration: 6 months

PHASE 3: DRY DOCK — COMPLETE VESSEL TESTING (TRL 6→7)

3.1 Current Injection — Complete Vessel

Test Configuration: - Vessel in dry dock (or floating at dock with shore power disconnected and gangway isolated) - 4 × 1,000 A current injection electrodes placed at cardinal points 100 m from vessel in surrounding water (or ground for dry dock) - Injection waveform: 0.001-1 Hz, 1-500 A per electrode (total 4-2,000 A injected) - 200+ measurement points: Rogowski coils at all BFRP/steel interfaces, voltage sensors bow-to-stern and port-to-starboard, magnetometers at 10 locations on each deck

Acceptance Criteria: - Hull-induced current (integrated across all measurement points) <0.03% of total injected current - No individual measurement point exceeds 0.1% of injected current - All dielectric joints (BFRP/steel interfaces) carry <10 mA at maximum injection - PMMA POF data network: zero induced current, zero BER increase during injection - LiFePO₄ battery BMS: no spurious protective trips triggered by induced currents - FOG navigation: heading drift <0.1° during maximum injection

Cost: \$1.2M; Duration: 2 weeks (test execution), 2 months (preparation + analysis)

3.2 System-by-System Verification

System	Test	Method	Acceptance
Emergency Propulsion (piezoelectric)	Full power test at dock	Run emergency propulsion at rated power (2,000 HP if available; otherwise diesel-electric with dielectric isolation) for 1 hour	All electrical parameters within spec; motor-to-hull resistance >10 ⁸ Ω; zero abnormal vibration
Navigation (FOG + POF)	Accuracy comparison vs. GPS over 24 hours at dock	Compare FOG heading, roll, pitch with GPS/INS reference	Heading error <0.1° sec lat (standard for marine FOG)
Fire Suppression (dielectric pumps)	Flow test at each of 5 fire zones	Activate fire pumps; measure flow at each hydrant; run for 1 hour continuous	Flow >spec per fire plan; pump motor temperature within limits; dielectric isolation maintained
Emergency Communications (Protonic)	12-node message relay test across vessel	Send 100 test messages between bridge, engine room, and 10 passenger deck nodes	>99% message delivery; latency <5 seconds per hop; zero messages lost
Hotel Load (LiFePO ₄ battery)	Full charge/discharge cycle	Charge battery from shore power; discharge at 5 MW hotel load for 4 hours (simulated GIC event blackout duration)	Battery temperature within limits; BMS no faults; capacity >95% of rated

Entertainment Systems (POF network)	Bandwidth test to 100 representative cabins	Stream 4K video to 100 cabins simultaneously via POF backbone	>95% of cabins report zero buffering; bandwidth per cabin >25 Mbps
Watertight Integrity	Hose test per SOLAS II-1	Spray all doors, hatches, cable penetrations with 12 mm nozzle at 10 m distance	No water ingress through any penetration

3.3 Safety Systems — Emergency Drills

Drill	Scenario	Duration	Acceptance
GIC Blackout Drill	All bridge electronics “fail” (simulated); crew must navigate on FOG backup; switch hotel load to LiFePO ₄ ; activate Protonic emergency communications; launch lifeboats (simulated)	2 hours	All systems transfer within SOLAS-required time limits; crew proficiency rated >80% by independent assessor
Fire in Battery Room	Simulated LiFePO ₄ thermal event; crew must isolate, suppress, and evacuate adjacent zones	1 hour	Fire team response time <5 minutes; suppression effective within 10 minutes; no simulated casualties
Casualty Evacuation	50 simulated casualties (passenger + crew) distributed across 5 decks; crew must locate, triage, and evacuate to muster stations	2 hours	All casualties located within 30 minutes; all evacuated within 2 hours; triage accuracy >90%

Phase 3 Cost: \$3.8M (inclusive of crew training, independent assessors, instrumentation rental);
Duration: 3 months (dry dock test window)

PHASE 4: SEA TRIALS (TRL 7→8)

4.1 Sea Trial Test Plan

Test	Location	Duration	Method	Acceptance
Speed/Power Trial	Open ocean, calm conditions (Sea State <3)	4 hours	Standardized speed trial per ITTC Recommended Procedures; measure shaft power at 50%, 75%, 90%, 100% MCR	Power within 5% of pre-retrofit baseline (MXene coating and BFRP structural weight — verify no drag penalty or power increase >2%)
Endurance Trial	Coastal transit, 24 hours	24 hours	Continuous operation at service speed; all systems monitored	Zero EMI-related anomalies; all GIC sensors within background; battery system nominal
Maneuvering Trial	Designated sea trial area	4 hours	Turning circle, zig-zag, crash stop per IMO resolution MSC.137(76)	Maneuvering characteristics within IMO standards; FOG navigation stable during all maneuvers
GIC Simulation at	Open ocean, >50 nm		Towed electrode array behind vessel; inject 0.001–1 Hz current into seawater; measure	Hull-induced current <0.03% of injected;

Sea	offshore	8 hours	hull response (similar to dry dock test but in actual operating environment — most realistic GIC simulation possible)	all systems nominal; navigation unaffected
Heavy Weather Trial	Open ocean, Sea State 5–6 (if available)	4 hours	Operate at reduced speed in heavy seas; monitor structural response of BFRP stiffeners via FBG strain gauges	BFRP strain within design limits; no structural acoustic emission (indicating no cracking); MXene coating adhesion maintained (visual inspection at next port)
Anchoring Trial	Designated anchorage, 30 m depth	2 hours	Deploy and retrieve both anchors; verify windlass current isolation	Windlass motor-to-hull resistance >10 ⁸ Ω; anchor chain isolated from hull (ceramic fairlead inserts verified)

4.2 Continuous Monitoring During Sea Trials

Parameter	Sensors	Data Volume (24 hours)
Hull voltage (bow–stern, port–starboard)	4 voltage sensors, 1 Hz	345,600 samples/sensor
GIC at BFRP/steel interfaces	200 Rogowski coils, 1 Hz	17.3 million samples
POF data network BER	OTDR monitoring at 6-minute intervals	240 measurements
Battery BMS telemetry	BMS CAN bus data, 1 Hz	86,400 samples
Navigation (FOG vs. GPS)	FOG + GPS comparison at 10 Hz	864,000 samples
Structural strain (FBG)	50 FBG sensors at 100 Hz	432 million samples
Engine/propulsion parameters	Engine room data logger, 0.1 Hz	8,640 samples
TOTAL		~500 GB per 24 hours

Phase 4 Cost: \$4.5M (ship time, fuel, crew, equipment); Duration: 2 weeks at sea + 2 months data analysis

PHASE 5: IN-SERVICE MONITORING (TRL 8→9)

5.1 12-Month Commercial Operation

After sea trials, the Charlemagne pilot vessel enters commercial service with all 200+ GIC sensors continuously monitoring. Key metrics tracked:

Metric	Target
GIC events detected (Kp ≥5 storm days)	Correlate with NOAA SWPC data; verify <0.03% induced current during actual geomagnetic storms
System availability (all hardened systems)	>99.9% uptime
Passenger/crew safety incidents attributable to EM systems	Zero
Insurance claims (compare to pre-retrofit baseline)	Documented reduction
Maintenance events on hardened systems	<2 per 1,000 operating hours
Passenger satisfaction (cruise surveys)	No negative comments related to EM hardening; <1% of passengers aware of hardening (transparent protection)

TARGET VERIFICATION — 0.03% INDUCED CURRENT

Verification Methodology

The Charlemagne design target of <0.03% induced current is verified through:

- 1. **Component Level:** Each of 200+ BFRP/steel interfaces individually verified >10⁸ Ω DC resistance (Phase 1)
- 2. **Sub-Scale:** 10 m hull section current injection shows <0.03% (Phase 2)
- 3. **Dry Dock:** Full vessel current injection shows <0.03% (Phase 3) — PRIMARY VERIFICATION
- 4. **Sea Trials:** At-sea GIC simulation confirms dry dock results under operational conditions (Phase 4)
- 5. **In-Service:** During actual geomagnetic storms, continuous monitoring confirms sustained <0.03% (Phase 5)

Metric Definition: Hull-Induced Current Ratio = (Total current measured across all BFRP/steel interface Rogowski coils) / (Total current injected into water) × 100%. Target <0.03% means >99.97% of injected current is blocked or diverted around the hull by MXene coating + BFRP isolation.

CHARLEMAGNE TESTING BUDGET SUMMARY

Phase	Duration	Cost
Phase 1: Component Validation	12 months	\$1.2M
Phase 2: Sub-Scale Hull Section	6 months	\$1.8M
Phase 3: Dry Dock Vessel Testing	3 months	\$3.8M
Phase 4: Sea Trials	2 months (at sea)	\$4.5M
Phase 5: In-Service Monitoring	12 months	\$1.8M
TOTAL	35 months	\$13.1M

Plus: \$47M for the vessel retrofit itself (not a testing cost — the vessel is a commercial asset after testing).

“A cruise ship is the most complex machine most humans will ever set foot on. Testing it for electromagnetic hardening is not about one measurement — it is about 200 simultaneous measurements at every conductive interface, verified at the component level, the sub-scale level, the dry dock level, at sea, and in service. When 3,000 passengers board a Charlemagne-hardened vessel, they are trusting their lives to a system that has been tested to the same standard as a military vessel entering a combat zone — because a Carrington Event is a battlefield, and the enemy is the Sun.” — Captain Meridia Vaun (CHARLEMAGNE)

Document generated 2026-07-15. All testing phases are contingent on securing a cruise line partner for the pilot vessel (see ROAD-09). Phase 1 (component validation) begins Q4 2026 upon MXene production at 10 kg/day. Phase 2 (hull section) requires dry dock access — target Q3 2027. Phase 3 (full vessel dry dock) tied to partner vessel’s scheduled dry dock window (target Q1 2028). ABS/DNV surveyors to witness all Phase 3 and Phase 4 testing.